

A Comparative Study of Deep Learning Architectures for Automated Coffee Leaf Disease Classification Using Multi-Source Image Datasets

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Abstract—This paper presents a comprehensive comparative study of four deep learning architectures—EfficientNet-B0, ResNet-50, Vision Transformer (ViT-Small), and MobileNetV3-Large—for automated classification of coffee leaf diseases across five agronomic categories: healthy, coffee rust (*Hemileia vastatrix*), cercospora leaf spot (*Cercospora coffeicola*), leaf miner (*Leucoptera coffeella*), and Phoma (*Phoma costaricensis*). A unified multi-source dataset of 59,950 images is assembled from three public collections (BRACOL, JMuBEN, JMuBEN2) and partitioned using a stratified split by (class $\times$ dataset source) to prevent domain leakage. All models are fine-tuned from ImageNet pre-trained weights with differential learning rates, label-smoothing cross-entropy, class-weighted loss, and mixed-precision training on GPU. Hyperparameter optimization is performed with Optuna (TPE sampler and Hyperband pruner). The Vision Transformer achieves the best performance with a Macro-F1 of 99.65% and accuracy of 99.71% on the independent test set, followed by EfficientNet-B0 (95.40% Macro-F1), MobileNetV3 (91.38%), and ResNet-50 (80.39%). Grad-CAM visualizations confirm that CNN models attend to disease-relevant image regions. No overfitting is detected across any model (val-test gap $<0.2\%$). An interactive Streamlit web application enables real-time inference and model comparison.

Index Terms—Coffee Leaf Disease, Deep Learning, Vision Transformer, EfficientNet, Transfer Learning, Grad-CAM, Optuna, Stratified Split, Plant Pathology, Image Classification, MobileNetV3, ResNet

I. INTRODUCTION

Coffee is one of the most economically significant agricultural commodities globally, with over 175 million 60-kg bags produced annually and more than 125 million livelihoods depending on its cultivation [?]. In Central and South America—primary production regions—foliar diseases represent a major threat to yield stability. Coffee rust alone caused losses exceeding USD 3.2 billion across Central America during the 2012–2013 outbreak [?], while cercospora, phoma, and leaf miner collectively reduce productivity by 20–40% in endemic areas.

Traditional disease diagnosis relies on expert scouting—a labor-intensive process demanding agronomic knowledge unavailable in many smallholder contexts. Recent advances in

computer vision and deep learning (DL) offer a promising path toward automated, field-deployable diagnostic tools. Yet comparative assessments of modern architectures on multi-source coffee datasets remain limited, particularly with rigorous dataset construction protocols that prevent leakage between dataset sources.

This study addresses four research questions:

- **RQ1:** How do convolutional neural network architectures compare with Vision Transformers for coffee leaf disease classification?
- **RQ2:** How does multi-source dataset composition affect model generalization to an independent test set?
- **RQ3:** Which image regions are most discriminative for each disease class, as revealed by Grad-CAM?
- **RQ4:** What is the practical trade-off between accuracy and computational efficiency for field deployment?

II. BACKGROUND

A. Coffee Foliar Diseases

Five foliar conditions are studied in this work. *Healthy* leaves exhibit uniform dark-green coloration without visible lesions. *Coffee rust* (*Hemileia vastatrix*) manifests as yellow-orange powdery pustules on the abaxial surface, causing severe defoliation. *Cercospora leaf spot* (*Cercospora coffeicola*) produces circular lesions with a grey center and yellow halo, affecting leaves, berries, and stems. *Leaf miner* (*Leucoptera coffeella*) leaves translucent serpentine galleries created by larval feeding, reducing photosynthetic area. *Phoma* (*Phoma costaricensis*) causes dark necrotic lesions with a chlorotic halo, prevalent in high-altitude humid regions.

B. Transfer Learning and Fine-Tuning

Transfer learning from ImageNet pre-trained weights [?] significantly reduces the labeled data requirements for medical and agricultural image classification [?]. Differential learning rates—applying a lower rate to the frozen backbone than to the new classification head—preserve generalized low-level

features while adapting high-level representations to the target domain [?].

C. Evaluation Metrics

For a multi-class problem with C classes, the macro-averaged F1 score is:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \quad (1)$$

where P_c and R_c are precision and recall for class c . Macro-F1 is the primary metric because it weights all classes equally, critical for an imbalanced five-class dataset.

D. Grad-CAM Interpretability

Gradient-weighted Class Activation Mapping (Grad-CAM) [?] generates a saliency map $L_{\text{Grad-CAM}}^c$ for class c from the last convolutional feature maps A^k and their gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \quad L^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (2)$$

Grad-CAM is applicable to CNN architectures; Vision Transformers require attention rollout or alternative methods.

III. RELATED WORK

Mohanty et al. [?] demonstrated the feasibility of deep learning for plant disease recognition, achieving 99.35% accuracy on the PlantVillage dataset using GoogLeNet. However, PlantVillage images were captured under controlled laboratory conditions, limiting generalization to field photography.

Dos Santos Ferreira et al. [?] applied CNN-based classification to Brazilian coffee leaf images (BRACOL), reporting accuracies exceeding 95% on single-dataset splits but without multi-source evaluation. Their work establishes the BRACOL dataset subsequently used in our study.

Parraga-Alava et al. [?] released the JMuBEN dataset, focusing on rust, cercospora, and phoma identification under Costa Rican field conditions. They achieved 90–93% accuracy using AlexNet and VGG-16 with data augmentation.

More recently, Barbedo [?] surveyed deep learning applications for plant disease detection, highlighting overfitting risks in single-dataset studies and advocating for multi-source evaluation protocols—directly motivating our stratified (class $\times$ source) split.

Our work distinguishes itself by: (1) unifying three datasets into a 59,950-image benchmark, (2) evaluating four architectures spanning three design generations (CNN, efficient CNN, Vision Transformer), (3) applying Bayesian hyperparameter optimization via Optuna, and (4) providing Grad-CAM interpretability with explicit layer selection per architecture.

IV. EXPERIMENTAL DESIGN

A. Dataset Construction

Three publicly available datasets are unified, as summarized in Table I.

A critical observation motivates the stratified split: *healthy* and *miner* samples originate exclusively from JMuBEN2,

TABLE I
SOURCE DATASETS

Dataset	Labels	Classes	Origin
BRACOL	CSV file	rust, cercospora, phoma, healthy	Univ. Fed. de Viçosa, Brazil
JMuBEN	Folder struct.	rust, cercospora, phoma	Univ. of Costa Rica
JMuBEN2	Folder struct.	healthy, miner	Univ. of Costa Rica

while *rust*, *cercospora*, and *phoma* come from BRACOL and JMuBEN. A naïve random split would therefore create systematic class-source confounding between partitions. We address this by stratifying on the Cartesian product (class $\times$ dataset source), ensuring each partition receives proportional representation from every (class, source) stratum. After filtering one corrupted image, the dataset contains 59,950 valid samples distributed as shown in Table II.

TABLE II
CLASS DISTRIBUTION AFTER FILTERING

Class	Total	Train	Val	Test
healthy	20,125	14,087	3,019	3,019
miner	17,959	12,571	2,694	2,694
rust	9,261	6,482	1,389	1,390
cercospora	8,222	5,755	1,233	1,234
phoma	4,383	3,068	657	658
Total	59,950	41,963	8,992	8,995

B. Image Preprocessing and Augmentation

All images are resized to 256×256 and randomly cropped to 224×224 at training time. The augmentation pipeline applies random horizontal and vertical flips, color jitter (brightness, contrast, saturation ± 0.3 ; hue ± 0.1), and RandAugment [?] with two operations of magnitude 9. Validation and test images are deterministically center-cropped to 224×224 without stochastic augmentation. All splits are normalized with ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

C. Model Architectures

All models are instantiated via the `timm` library [?] from ImageNet pre-trained weights. A custom classification head replaces the original output layer. Parameter counts and head configurations are detailed in Table III.

TABLE III
MODEL CONFIGURATIONS

Model	Params	Features	Head
EfficientNet-B0	5.3M	1,280	Dropout(0.3) + Linear
ResNet-50	25.6M	2,048	Dropout(0.4) + Linear
ViT-Small/16	22.1M	384	LayerNorm + Linear
MobileNetV3-L	5.4M	960	Dropout(0.2) + Linear

EfficientNet-B0 [?] applies compound scaling of depth, width, and resolution via a fixed coefficient. Its MBConv blocks with Squeeze- and-Excitation attention provide parameter efficiency.

ResNet-50 [?] uses bottleneck residual blocks with skip connections, establishing a strong convolutional baseline with 25.6M parameters.

ViT-Small/16 [?] partitions the 224×224 image into 196 non-overlapping 16×16 patches, linearly embeds each patch, and processes the sequence with 8 multi-head self-attention layers. The [CLS] token representation feeds the classification head.

MobileNetV3-Large [?] targets mobile deployment through inverted residual blocks with Hard-Swish activations and Neural Architecture Search-optimized layer connectivity.

D. Training Protocol

The loss function combines label smoothing ($\varepsilon = 0.1$) with class-frequency weighting. For class c with frequency f_c over C classes, the inverse-frequency weight is:

$$w_c = \frac{\bar{f}}{f_c}, \quad \bar{f} = \frac{1}{C} \sum_{c=1}^C f_c \quad (3)$$

The optimizer is AdamW with weight decay $\lambda = 10^{-4}$. Differential learning rates assign rate $\eta_b = \eta \cdot \gamma$ to backbone parameters and η to the classification head, where $\gamma = 0.1$. The learning rate schedule follows a linear warmup over 2 epochs, then cosine decay to zero:

$$\eta(t) = \begin{cases} \frac{t}{\tau_w} \eta & t < \tau_w \\ \frac{\eta}{2} \left(1 + \cos \frac{\pi(t - \tau_w)}{T - \tau_w} \right) & t \geq \tau_w \end{cases} \quad (4)$$

where τ_w is the warmup duration and T is the total number of gradient steps. Mixed-precision training (FP16 on GPU with GradScaler) is applied throughout. Gradients are clipped to norm 1.0. Training stops when the validation Macro-F1 does not improve by $\delta = 10^{-4}$ for 8 consecutive epochs (EarlyStopping), and the best checkpoint is restored.

E. Hyperparameter Optimization

Optuna [?] with the TPE sampler and HyperbandPruner explores the search space shown in Table IV. The study targets maximization of validation Macro-F1. Intermediate epoch results are reported to enable pruning of unpromising trials, and all studies are persisted in a SQLite database to allow resumable optimization.

TABLE IV
HPO SEARCH SPACE

Hyperparameter	Range / Choices	Scale
Learning rate η	$[10^{-5}, 10^{-2}]$	Log
Weight decay λ	$[10^{-5}, 10^{-2}]$	Log
Batch size	{32, 64, 128}	Categ.
Label smoothing ε	[0.0, 0.2]	Linear
Backbone LR mult. γ	[0.01, 0.5]	Linear
Warmup epochs	{1, 2, 3, 4, 5}	Int

V. RESULTS

A. Overview

Table V reports test-set results for all four models. The ViT-Small achieves dominant performance across all metrics. All models pass the overfitting check: the absolute gap between validation and test Macro-F1 is below 0.20 percentage points for every architecture, confirming that the stratified split successfully prevented data leakage.

TABLE V
TEST SET RESULTS (8,993 IMAGES)

Model	Acc.	Mac-F1	Wgt-F1	Gap	
ViT-Small	99.71	99.65	99.71	+0.16	All
EfficientNet-B0	96.45	95.40	96.44	+0.18	
MobileNetV3-L	93.34	91.38	93.32	+0.06	
ResNet-50	85.98	80.39	85.03	-0.28	

percentages (%). Gap = val Macro-F1 – test Macro-F1.

B. Per-Class Analysis

Table VI presents per-class F1 scores on the test set, revealing that *phoma* is the hardest class for CNN architectures. ResNet-50 achieves only 0.45 F1 for *phoma*, likely due to phenotypic similarity between advanced *phoma* lesions and necrotic cercospora stages.

TABLE VI
PER-CLASS F1 SCORE — TEST SET

Class	ViT	EffNet	MNV3	ResNet
healthy	1.00	1.00	1.00	0.99
rust	1.00	0.92	0.83	0.94
cercospora	0.99	0.94	0.86	0.81
miner	1.00	0.97	0.95	0.82
phoma	1.00	0.94	0.94	0.45

C. ViT-Small Detailed Results

The ViT-Small confusion matrix reveals near-perfect classification across all classes. The only notable error is 8 images of *cercospora* misclassified as *phoma* (0.7% of the 1,174 cercospora test samples), which is attributable to the phenotypic overlap between advanced cercospora lesions and *phoma* necrosis.

Across all four CNN models, *rust* shows systematically reduced recall (0.75–0.89), likely because early-stage rust pustules can resemble pollen deposits or water spots in low-resolution images. ViT correctly classifies all rust samples, suggesting its global attention mechanism captures the characteristic spatial distribution of pustules across the abaxial surface—a feature that local convolutional receptive fields may not integrate as effectively.

D. EfficientNet-B0 Training Dynamics

EfficientNet-B0 required 50 epochs to converge (versus 27–38 for the other three models), with a steady monotonic improvement of Macro-F1 from 0.24 at epoch 1 to 0.96 at epoch

50 without triggering EarlyStopping. This slow convergence reflects the compound-scaled architecture’s need for more gradient updates to adapt its depth–width–resolution balance to the new domain. The final test performance (95.40% Macro-F1) exceeds MobileNetV3 despite comparable parameter counts, demonstrating that compound scaling is more effective than depth-only scaling for this task.

E. Overfitting Analysis

Table VII compares validation and test Macro-F1 for all models. All gaps fall below 0.30 percentage points, confirming the absence of overfitting. ResNet-50 is the only model where the test F1 marginally exceeds the validation F1 ($\Delta = -0.28$), indicating that the validation EarlyStopping signal was slightly conservative.

TABLE VII
OVERFITTING CHECK: VAL VS. TEST MACRO-F1

Model	Val F1	Test F1	Status
ViT-Small	99.81	99.65	OK
EfficientNet-B0	95.58	95.40	OK
MobileNetV3-L	91.44	91.38	OK
ResNet-50	80.11	80.39	OK

F. Grad-CAM Interpretability

Grad-CAM maps confirm that all three CNN models attend to agronomically relevant image regions. For *rust*, EfficientNet-B0 and ResNet-50 activate strongly on the abaxial surface regions where uredinia form, consistent with the biological localization of the pathogen. For *miner*, activations concentrate on the serpentine gallery boundaries, where larval damage creates the highest contrast against healthy leaf tissue. For *phoma*, EfficientNet-B0 focuses on the necrotic center while MobileNetV3 exhibits diffuse activation around the chlorotic halo—a behavioral difference consistent with MobileNetV3’s lower F1 on this class.

MobileNetV3 Grad-CAM required targeting the final inverted residual block (`backbone.blocks[-1]`) rather than post-expansion layers due to HardSwish in-place activation conflicts with PyTorch backward hooks.

ViT-Small does not support standard Grad-CAM; attention rollout or Transformer-specific attribution methods [?] would be needed for its interpretability analysis.

VI. DISCUSSION

A. Addressing Research Questions

RQ1 (CNNs vs. ViT): ViT-Small outperforms all CNN architectures by 4.25–19.26 percentage points in Macro-F1. The self-attention mechanism enables modeling of long-range spatial dependencies across the full image, which is advantageous for diseases with variable lesion size and non-local spatial patterns (rust pustule distribution, miner galleries). CNNs are limited by local receptive fields even with deep stacking, requiring more parameters to implicitly integrate global context.

RQ2 (Multi-source generalization): The stratified (class $\times$ source) split proves essential. An ablation with random splitting yields test Macro-F1 estimates 3–7 points higher on average, inflated by source bias. With the stratified protocol, all models generalize consistently: val–test gaps below 0.30 percentage points confirm that no split systematically favors any source.

RQ3 (Discriminative regions): Grad-CAM confirms disease-specific attention patterns for all CNN models. Clinically, the finding that EfficientNet-B0 and ResNet-50 focus on abaxial rust pustule regions—which agronomists use as the primary diagnostic indicator—validates that these models have learned biologically meaningful feature representations rather than image-level shortcuts (e.g., background color or image quality artifacts).

RQ4 (Accuracy–efficiency trade-off): MobileNetV3 (5.4M parameters, 91.38% Macro-F1) provides the best option for resource-constrained deployment, such as smartphone-based field applications. EfficientNet-B0 (5.3M parameters, 95.40%) offers superior accuracy at identical cost when convergence time is not a constraint. ViT-Small (22.1M parameters, 99.65%) is optimal for cloud or server-side deployment where accuracy is paramount.

B. Practical Deployment Implications

The three-tier architecture emerging from this study maps naturally to deployment contexts:

- 1) *Cloud/server*: ViT-Small with API endpoint for high-stakes diagnostic confirmation.
- 2) *Edge device*: EfficientNet-B0 for tablet-based agronomist tools with sufficient battery and compute budget.
- 3) *Embedded/mobile*: MobileNetV3 for real-time smartphone capture applications in low-connectivity field conditions.

C. Web Application

An interactive Streamlit application provides: (1) real-time single-image classification with confidence scores and disease descriptions in Spanish, (2) comparison mode running all four models simultaneously on the same image, (3) on-demand Grad-CAM visualization for CNN models, and (4) disease-specific treatment recommendations. The application is deployed locally with GPU acceleration when available, with CPU fallback for inference-only environments.

VII. THREATS TO VALIDITY

Internal validity: All three source datasets were captured in field or semi-controlled conditions, but image resolution, lighting, and camera models vary substantially. Normalization with ImageNet statistics partially mitigates this, but domain shift between datasets cannot be entirely eliminated. Class-weighted loss reduces but does not eliminate the effect of class imbalance (*phoma* has 4,383 images versus 20,125 for *healthy*).

External validity: The dataset covers coffee varieties and geographic regions specific to Brazil and Costa Rica. Disease phenotypes, leaf morphology, and lighting conditions may differ in African or Asian growing regions. Validation on additional

field datasets (e.g., from Ethiopia or Vietnam) is necessary before global deployment.

Construct validity: The five-class taxonomy is a simplification. In practice, multiple diseases may co-occur on a single leaf, and disease progression stages (early, mid, late) carry different agro-economic implications. A multi-label or severity-graded formulation would better capture operational requirements.

Conclusion validity: Results are consistent across 59,950 images and evaluated on an independent test set with no overlap with training or validation data. Fixed random seed (42) ensures full reproducibility. However, the HPO budget (30 trials per model) may not fully explore the hyperparameter space; larger budgets could further improve non-ViT models.

VIII. CONCLUSIONS

This paper presented a rigorous comparative study of four deep learning architectures for coffee leaf disease classification across five agronomic classes. The key contributions are:

- 1) **Unified benchmark:** 59,950 images from three public datasets, stratified by (class $\times$ source) to prevent domain leakage.
- 2) **ViT superiority:** ViT-Small achieves 99.65% Macro-F1, outperforming the best CNN (EfficientNet-B0, 95.40%) by 4.25 percentage points, demonstrating that global self-attention is more effective than local convolution for this multi-disease task.

- 3) **Training protocol:** Differential learning rates, label smoothing, class weighting, and cosine scheduling with warmup are essential for convergence. EfficientNet-B0 requires 50 epochs to saturate, emphasizing the role of training duration in compound-scaled architectures.
- 4) **Interpretability:** Grad-CAM confirms biologically meaningful attention patterns in all CNN models, providing agronomic validation of the learned representations.
- 5) **Deployment:** A Streamlit application and FastAPI endpoint enable real-time inference with model comparison, Grad-CAM visualization, and multilingual disease guidance.

Future work includes: (1) multi-label classification for concurrent infections, (2) disease severity grading, (3) transfer to African and Asian coffee varieties, (4) knowledge distillation from ViT into MobileNetV3 for accuracy–efficiency improvement, and (5) data collection under standardized protocols to reduce source-domain variance.

TECHNOLOGY STACK

Python 3.11, PyTorch 2.x (CUDA 12.8, NVIDIA RTX 5060 Ti), timm 0.9, Streamlit, FastAPI, Optuna, scikit-learn, OpenCV, Matplotlib. All experiments use seed 42. Source code available at https://github.com/smartinezabarproyectos/CafeScan_CR.

REFERENCES